LOGISTICSOLVER 2026 · FULLY PUBLIC EVIDENCE

Fee cliffs.
Better schedules.
Priced resilience.

PortPlan — container pickup, unloading and empty-return planning under public Korean tariff rules.

ENGLISH PRESENTATION DRAFT · 14 SLIDES · 12 OCT 2026 KST

Public tariff examples, controlled scheduling experiments and real public gate observations.

One extra day can be expensive.

₩318,500

PNC storage · 20 ft · day 14

₩500,500

PNC storage · 20 ft · day 15

+₩182,000

The extra day crosses a retroactive tier

PNC applies the highest reached rate to **all chargeable days beyond the four free days**. A smooth daily-cost approximation can miss this jump.

₩318,500 = 10 × 31,850; ₩500,500 = 11 × 45,500. Separate Maersk D&D continues until empty return.

Source: PNC Standard Tariff 2026, p.4; Maersk PNC storage notice. Standard dry only, taxes/contracts excluded.

Plan the full container lifecycle.

Release → **Pickup** → **Unload** → **Empty return**

Decisions and constraints

- One pickup, unloading start and return date per box
- Daily movement, unloading and retained-box capacity
- Release dates and hard unloading deadlines

What the model excludes

- Road routes, driver hours and appointment slots
- Negotiated contracts, taxes and transport prices
- Online information and already-executed actions

Objective: terminal storage + carrier D&D. “Optimal” means optimal within this stated static daily model.

Separate rules, assumptions and observations.

Evidence	What it supports	What it cannot prove
PNC 2026 official tariff	Retroactive storage calculation	Every shipment's negotiated bill
Maersk Korea official example	10 inclusive free days; first-band charge	Cross-band accumulation is still an interpretation
58,171 public Gwangyang gate records	Held-out gate-duration estimation	PNC orders, return chains or fee savings
Generated order scenarios	Controlled, reproducible comparisons	Realized enterprise savings

Two published numerical examples and eleven derived boundary cases are explicitly separated.

Sources and applicability: source_registry.json; source-audit.md; rules-status-v2.md.

Test on a later month.

9,152

June gate visits held out for testing

16.47 min

Mean absolute error · hour-based median

2.42%

Lower error than fixed global median

January–April trains the models (41,039 visits). May selects among three declared methods (7,980 visits). June evaluates without refitting.

Only entry hour and weekday are available to the estimator. Baseline MAE: 16.88 minutes; P90 absolute error: 35 minutes.

> The improvement is modest. These are Gwangyang gate durations, not PNC fee savings or road-travel predictions.

An auditable local application. No algorithmic first claim.

Prior work already exists

- Jeong et al. (2020): devanning and detention
- Jeong et al. (2026): uncertain D&D decisions
- Drayage MILP, release/due dates and empty-stock coordination
- PortPro: pickup/return appointment matching

Contribution to test

- Encode PNC's exact 2026 fee cliff
- Audit the complete pickup/return cost chain
- Expose sensitivity, failures and buffer premiums
- Publish data, code and independent checks

Nine research works and three products reviewed at disclosed depths. A focused search does not establish global novelty.

References: prior-art-v2.md. DOI 10.1016/j.tre.2020.102055; 10.1111/itor.13622.

Keep the original failure visible.

Original PLOS code

Six fixed runs fail because the archive omits the file **distance2** requested by its algorithm.

The source files and algorithm functions remain unchanged.

Explicit supplemental assumption

Six runs with **distance2 = supplied distance** return results at a reduced, declared budget.

Node and timestamp counts disagree in routes; complete feasibility and costs remain unverified.

> A separate minute-level routing model. No claim that PortPlan outperforms it; no cross-objective cost leaderboard.

Freeze the questions before the run.

48

New seed scenarios · 16 / 50 boxes

3

Backlog / new arrivals / batch release

6

Policies under identical inputs and constraints

Stronger internal comparator: earliest-deadline rule, 12 randomized permutations and 80 relocate/swap attempts, all with a complete greedy decoder.

Transparent budgets: fixed search attempts versus a two-second MILP limit. Wall time is reported; budgets are not claimed equal.

Four 100-box controls and 56 sensitivity/ablation runs are separate. All cases retained.

Main protocol frozen in commit 1fb4678 before runs. Enhanced internal baseline, not a reproduced state-of-the-art algorithm.

Value is concentrated, not universal.

44/48

MILP complete feasible · 33 optimal / 11 incumbent

36/48

Stronger search complete feasible

15.74%

Lower incremental two-fee sum · 36 paired cases

Total two-fee reduction for those same pairs: **8.86%**. Nineteen improve; seventeen tie. Twelve cases lack a complete baseline and are not assigned invented savings.

> All 14 paired new-arrival cases cost zero under both methods. The value case is fee-exposed backlog and concentrated release.

Main experiment: synthetic inputs; four model-infeasible cases retained. Cost-weighted descriptive aggregate, not an expected business saving.

Check why a plan changes.

Deliberate model change	Extra cost under the same reference tariff	Interpretation limit
Smooth PNC as progressive tiers	₩185,200 / 2.22% across 6 pairs	Evidence from a small fixed panel
Optimize storage only	₩87,550 across 5 strict pairs	Exclude one unresolved secondary tie-break
Optimize D&D only	₩487,550 across 6 pairs	Entire difference comes from one case

Single-objective runs first minimize their own fee, then use remaining budget to minimize joint fees among ties. Unfinished tie handling stays visible.

Custom free days shift tariff tiers in this research model. That is not a verified contract-extension rule.

Sensitivity is paired scenario evidence, not proof of a universal mechanism or actual contractual exposure.

Keep the feasible plan when MILP times out.

100 boxes

New arrivals · tight capacity

Strong search found a complete zero-fee plan. MILP returned no incumbent within two seconds.

Decision-support correction

PortPlan recommends the lowest-cost complete audited candidate among the available policies.

It preserves each raw solver status and does not label a fallback globally optimal.

> A time limit is not an infeasibility proof. No incomplete plan enters the fee comparison.

Counterexample retained in scaleControls. This is a portfolio selection rule, not a new optimality guarantee.

A static optimum can be fragile.

16/18

Original plans invalid under fixed disruptions

18/18

Hindsight re-solves feasible

≠ online

Future disruptions were known in re-solves

Fixed tests reduced daily transport by two on days 3–5, extended every fourth box's unloading by one day, or combined both changes.

Six base scenarios are reused across disruptions. These are dependent stress cases, not a sampled field failure rate.

Independent audit uses actual processing durations and capacities, not cached completion fields.

Resilience has a measurable premium.

16 → 0

Invalid plans across the 18 specified checks

+32.29%

Total two-fee premium across 6 pairs

₩4,652,550

Additional modeled fees

Buffer rule: add one unloading day to every box and reserve two transport movements on every planning day. It does not select the disrupted boxes or dates.

Incremental-fee premium: 76.88% — a different denominator. Three new-arrival cases add no fees; backlog cases bear the premium.

> Exploratory study designed after observing failures. Valid only inside the stated duration/capacity envelope and daily model.

Make the decision inspectable.

Demonstration

1. Recalculate the official tariff examples.
2. Inspect real public gate-data holdout errors.
3. Compare six schedules and the buffer tradeoff.
4. Export sources, assumptions and results.

Submission scope

- All evidence is available online
- Official arithmetic and model experiments are distinct
- Public observations validate a gate-duration component
- Private contracts and realized savings are outside the claim

An auditable public-data project: exact fee rules, conditional scheduling value and an explicit price for resilience.